

Artificial Intelligence Ophthalmology Diagnosis Solutions



By Yuhui Sun, CEO, Visionary Intelligence Ltd.

» Visionary Intelligence Ltd. (Vistel)

Vistel is a strategic high-tech Chinese enterprise committed to the application of AI in eye healthcare via hands-on cooperation with ophthalmological teams of best Class-A hospitals including Peking Union Medical College Hospital, Beijing Tongren Hospital, Peking University Third Hospital, Beijing Hospital, Shanghai Eye Disease Prevention and Treatment Center, Zhongshan Hospital.



➤ Summary

Technology:

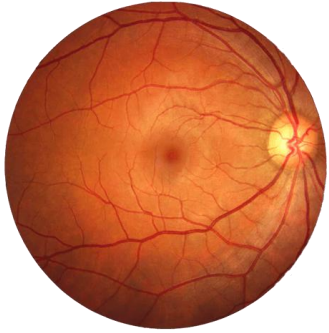
Solid deep learning AI technology roadmap, each one has 50K to 90K high quality labelled data and patent AI algorithm. There're 36 peer reviewed publications to prove the advancement.

Product Performance:

- World first & only AI software which detects up to 13 retinal disorders.
- DR AI software shows best clinical trial performance.
- Glaucoma AI software reduces false positive from 30%+ to 7.6%.
- It takes 10 second to get the AI diagnosis result.
- Widely accepted: 1000 medical institutes deployed so far
- 3M patients used in China.
- Won Poland DR screening program,
- used in several EU countries such as France, Spain, Germany, etc.

➤ Retinal Disorders: Glaucoma

healthy retina



Glaucoma

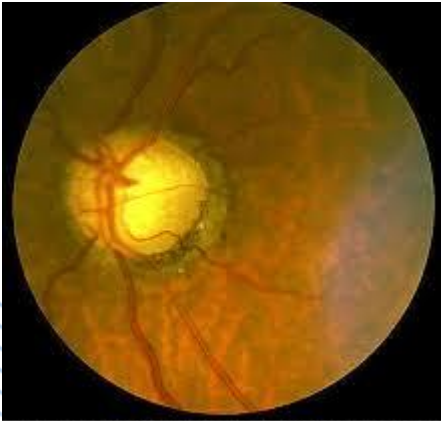


Fig. 23. A view of the fundus of the eye and of the retina in a patient who has advanced glaucoma.

The global prevalence of glaucoma for population aged 40-80 years is **3.54%** .

Source: Tham YC, Li X, Wong TY, Quigley HA, Aung T, Cheng CY. Global prevalence of glaucoma and projections of glaucoma burden through 2040: a systematic review and meta-analysis. *Ophthalmology*. 2014 Nov;121(11):2081-90. doi: 10.1016/j.ophtha.2014.05.013. Epub 2014 Jun 26. PMID: 24974815.

Glaucoma was the cause for blindness in **3.61 million people** or 8.4% of the 43.3 million blind people globally in 2020.

Source: Vision Loss Expert Group of the Global Burden of Disease Study., the GBD 2019 Blindness and Vision Impairment Collaborators. Global estimates on the number of people blind or visually impaired by glaucoma: A meta-analysis from 2000 to 2020. *Eye* **38**, 2036–2046 (2024). <https://doi.org/10.1038/s41433-024-02995-5>

Glaucoma checks recommendation: Every 1-2 years for adults 40-54, Every 1-3 years for adults 55-64, Every 1-2 years for adults 65+.

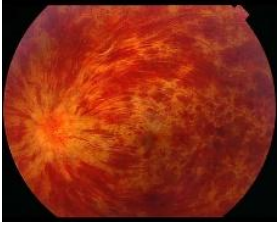
Source: <https://glaucoma.org/articles/the-importance-of-regular-eye-exams-to-detect-glaucoma>



GLAUCOMA
RESEARCH FOUNDATION

» Other Retinal Disorders

Retinal vein occlusion (RVO)



Retinal vein occlusion (RVO) is the second most common cause of vision loss due to retinal vascular disease. Population-based studies report a prevalence rate of 0.5-2.0% for branch RVO and 0.1-0.2% for central RVO.

Source: Laouri, Marianne & Chen, Er & Looman, M & Gallagher, Meghan. (2011). The burden of disease of retinal vein occlusion: Review of the literature. Eye (London, England). 25. 981-8. 10.1038/eye.2011.92.



Pathologic Myopia



The prevalence of pathologic myopia is only 1% to 19% in the low-to-moderate myopia (−3 D) population, but its prevalence is as high as **50% to 70% in the high myopia** (−6 D) population.

Source: Ohno-Matsui K, Wu P-C, Yamashiro K, et al. IMI Pathologic myopia. Invest Ophthalmol Vis Sci. 2021;62(5):5.



Epiretinal membranes (ERMs)



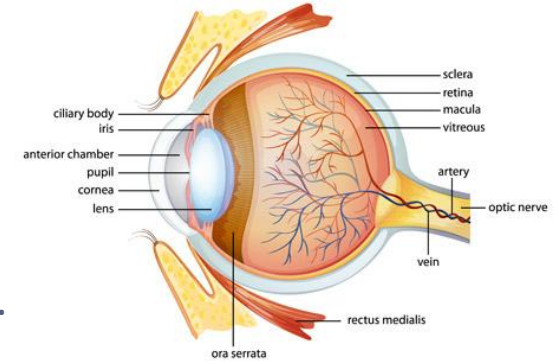
Epiretinal membranes (ERMs) can cause visual impairment, including blurred or distorted vision, and difficulty with fine details.

The prevalence of any **ERM was 28.9%**, of which 25.1% were CMR and 3.8% were PMF.

Source: Ng CH, Cheung N, Wang JJ, Islam AF, Kawasaki R, Meuer SM, Cotch MF, Klein BE, Klein R, Wong TY. Prevalence and risk factors for epiretinal membranes in a multi-ethnic United States population. Ophthalmology. 2011 Apr;118(4):694-9. doi: 10.1016/j.ophtha.2010.08.009. Epub 2010 Oct 29. PMID: 21035863; PMCID: PMC3070851.



➤ Challenges in Retinal Disorders Treatment



- **Early Diagnosis:**

- Many retinal diseases share similar symptoms, making early and accurate diagnosis difficult.

- **Treatment Burden:**

- Available treatments, such as anti-VEGF injections or surgery, can be burdensome for patients, requiring frequent appointments and potentially causing discomfort or side effects.

- **Limited Access to Specialized Care:**

- A shortage of retina specialists, especially in certain regions, can lead to delays in diagnosis and treatment, potentially worsening outcomes.

- **Impact on Quality of Life:**

- Retinal disorders can significantly affect daily activities, such as reading, driving, and navigating low-light environments, leading to frustration, social isolation, and dependence on others.

Early detection of retinal disorders is crucial for preserving vision, as many conditions worsen over time and can lead to irreversible damage if left untreated. Regular eye exams, especially retinal imaging, are essential for identifying these issues before symptoms become severe. **AI is the ONLY solution to do it in large scale due to ophthalmologist shortage.**



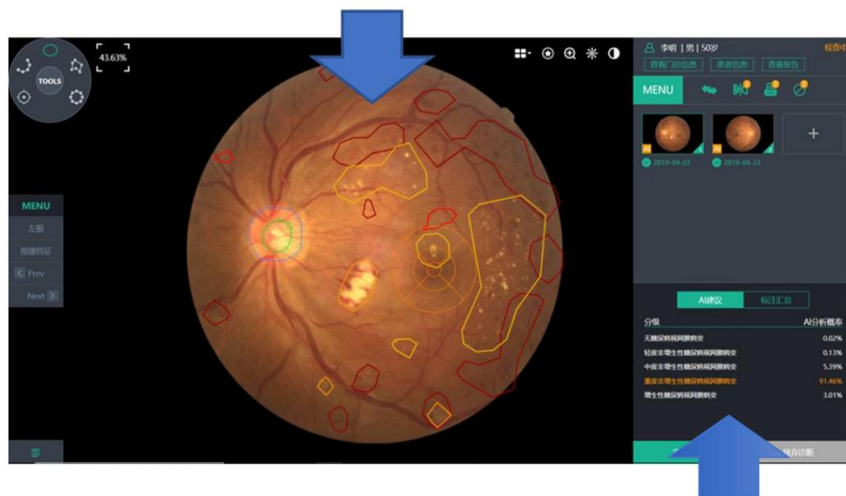
AI Product Matrix

- EyeWisdom AI-based Auxiliary Diagnosis System
- EyeWisdom DSS—— Auxiliary diagnosis system for DR
- EyeWisdom MCS——Auxiliary diagnosis system for multiple eye diseases
- EyeWisdom Glaucoma

» EyeWisdom DSS

AI-based Auxiliary Diagnosis System for DR

Automated annotation of lesions



Based on the International Clinical DR Classification System

EyeWisdom®DSS AI-assisted Diagnosis System can realize precise DR grading and statistical analysis of lesion annotations, and the analysis results can be obtained immediately within **10 seconds**. It is clinically applied to the large-scale screening and auxiliary diagnosis of DR, and has performed "early detection of eye diseases with a fundus image".

Dual Certification



NMPA



European
Union CE

Algorithm

An industrial leader in sensitivity under the same specificity conditions;

up to **92.96%**

EyeWisdom DSS Demo

The image displays two side-by-side screenshots of the EyeWisdom DSS (Decision Support System) interface, demonstrating its capabilities for retinal fundus image analysis.

Left Screenshot: Shows a retinal fundus image with various lesions highlighted by colored outlines. A callout box identifies a "Retinal hemorrhage". The sidebar on the right displays patient information: "DR Male 55 Years", "Exam ID: 202412250001", and "Waiting for process". A notification prompts the user to "Complete the outpatient info to improve the accuracy of AI diagnosis!". Below this, there are two small fundus images labeled 1 and 2, each with "Report" and "Mark" buttons. The "Suggestions" section shows "diabetic retinopathy" with a "severe NPDR" level. The "Fundus Analysis" section shows "diabetic retinopathy" with a "Highly Sus." (Highly Suspect) status.

Right Screenshot: Shows another retinal fundus image with a callout box identifying "Fibrous proliferation". The sidebar on the right displays patient information: "test patient Male 55 Years", "Exam ID: 202412250002", and "Waiting for process". It also features two small fundus images labeled 1 and 2, each with "Report" and "Mark" buttons. The "Suggestions" section shows "diabetic retinopathy" with a "PDR" (Proliferative Diabetic Retinopathy) level. The "Fundus Analysis" section shows "diabetic retinopathy" with a "Highly Sus." status, "rd" (retinal detachment) with a "Highly Sus." status, and "macular epiretinal membrane" with a "Per exclusi..." (Partially excluded) status. A "Generate Report" button is visible at the bottom.

EyeWisdom MCS

AI-based Auxiliary diagnosis system for multiple eye diseases

As the first AI-assisted diagnosis system of 10+ major blinding eye disorders in the world, it has the following features:

based on the AI analysis of a single color fundus photo of the posterior pole, it can judge whether there are abnormal lesions, and give the corresponding qualitative and quantitative auxiliary diagnosis analysis of fundus diseases, helping doctors to make accurate diagnoses.

Can detect the following retinal disorders: DR, RVO, wAMD, dAMD, Optic nerve atrophy, Macular Hole, Pathologic Myopia, Retinal detachment, Glaucoma, Central serous chorioretinopathy, the preretinal macular membrane)

Dual
certification

Only one of its kind that obtained
CE marking in the world
NMPA Class III medical device
approved in China, the first in the
country.



NMPA



European
Union CE

13 diseases

The only one in China
to have completed the clinical trials
of the 10+ fundus diseases
registered with NMPA

Patent

Innovative patented technology
Greatly raised product performance



Demo

23.23%

AI Feedback Download original image Display Analysis Hide annotation See enhanced image Zoom brightness/contrast

Wet AMD female 79 years old

Check Complete

Inspection number: 20230423...

View clinic information

2023 04 23

2023 04 23

Right Eye Left Eye Right Eye Left Eye

Diagnostic recommendations

Fundus structure and

Age-related macular degenera... Suspected

View Report

Microaneurysm

53.41%

TOOLS

MENU

Tags

Prev

Next

Cancel

Fibrous proliferation

Report template INTERNATIONAL

test patient Male 55 Years

Waiting for process

Exam ID: 202412250002

Exam details

2024-12-25

2024-12-25

Right Left Right Left

Report Mark Report Mark

DR Level

diabetic retinopathy

POR

Fundus Analysis

macular epiretinal membrane

Highly Sus...

diabetic retinopathy

Suspected

hypertensive retinopathy

Suspected

Macular Edema

Per exclusi...

Generate Report

46.1%

TOOLS

MENU

Tags

Prev

Next

Cancel

Report template INTERNATIONAL

test patient Male 55 Years

Waiting for process

Exam ID: 202412250003

Exam details

2024-12-25

2024-12-25

Right Left Right Left

Report Mark Report Mark

Suggestions

Lesions

+ Add Diagnosis

DR Level

diabetic retinopathy

no DR

Fundus Analysis

rd

Highly Sus...

hypertensive retinopathy

Highly Sus...

Generate Report

46.1%

TOOLS

MENU

Tags

Prev

Next

Cancel

Report template INTERNATIONAL

test patient Male 55 Years

Waiting for process

Exam ID: 202412250003

Exam details

2024-12-25

2024-12-25

Right Left Right Left

Report Mark Report Mark

Suggestions

Lesions

+ Add Diagnosis

DR Level

diabetic retinopathy

no DR

Fundus Analysis

rd

Highly Sus...

hypertensive retinopathy

Highly Sus...

Generate Report

13 Retinal Disorders which can be detected by our AI Software

Item	Retinal Disorders	Importance
1	Diabetic Retinopathy (DR)	The leading cause of blindness in the working-age population
2	Pathologic Myopia (PM)	Myopia is more and more popular in Asian countries
3, 4	Age-related Macular Degeneration (Dry/Wet)	The leading cause of blindness in the elderly
5	Retinal Vein Occlusion (RVO)	The retinal disease most likely to occur in patients with hypertension
6	Suspicious Glaucoma	The leading cause of irreversible blindness worldwide
7	Epiretinal Membrane (ERM)	One of the major macular diseases
8	Retinal Detachment (RD)	Common diseases requiring retinal surgical treatment
9	Macular Hole	One of the major macular diseases
10	Retinitis Pigmentosa (RP)	The most common hereditary retinal disease
11	Central serous chorioretinopathy (CSC)	One of the main retinal diseases in young patients
12	Optic atrophy	he final outcome of various optic nerve diseases
13	Retinal Artery Occlusion	One of the emergency retinal diseases

Peered Review Papers

An example of a paper published in Nature Digital Medicine - Capable of diagnosing 13 diseases

npj | digital medicine

www.nature.com/npjdigitalmed

Check for updates

ARTICLE OPEN

The performance of a deep learning system in assisting junior ophthalmologists in diagnosing 13 major fundus diseases: a prospective multi-center clinical trial

Bing Li^{1,2}, Huan Chen^{1,2}, Weihong Yu^{1,2}, Ming Zhang³, Fang Lu³, Jingxue Ma⁴, Yuhua Hao⁴, Xiaorong Li⁵, Bojie Hu⁵, Lijun Shen⁶, Jianbo Mao⁶, Xixi He^{7,8}, Hao Wang⁹, Dayong Ding⁹, Xirong Li¹⁰ and Youxin Chen^{1,2}✉

Artificial intelligence (AI)-based diagnostic systems have been reported to improve fundus disease screening in previous studies. This multicenter prospective self-controlled clinical trial aims to evaluate the diagnostic performance of a deep learning system (DLS) in assisting junior ophthalmologists in detecting 13 major fundus diseases. A total of 1493 fundus images from 748 patients were prospectively collected from five tertiary hospitals in China. Nine junior ophthalmologists were trained and annotated the images with or without the suggestions proposed by the DLS. The diagnostic performance was evaluated among three groups: DLS-assisted junior ophthalmologist group (test group), junior ophthalmologist group (control group) and DLS group. The diagnostic consistency was 84.9% (95%CI, 83.0% ~ 86.9%), 72.9% (95%CI, 70.3% ~ 75.6%) and 85.5% (95%CI, 83.5% ~ 87.4%) in the test group, control group and DLS group, respectively. With the help of the proposed DLS, the diagnostic consistency of junior ophthalmologists improved by approximately 12% (95% CI, 9.1% ~ 14.9%) with statistical significance ($P < 0.001$). For the detection of 13 diseases, the test group achieved significant higher sensitivities (72.2% ~ 100.0%) and comparable specificities (90.8% ~ 98.7%) comparing with the control group (sensitivities, 50% ~ 100%; specificities 96.7 ~ 99.8%). The DLS group presented similar performance to the test group in the detection of any fundus abnormality (sensitivity, 95.7%; specificity, 87.2%) and each of the 13 diseases (sensitivity, 83.3% ~ 100.0%; specificity, 89.0 ~ 98.0%). The proposed DLS provided a novel approach for the automatic detection of 13 major fundus diseases with high diagnostic consistency and assisted to improve the performance of junior ophthalmologists, resulting especially in reducing the risk of missed diagnoses. ClinicalTrials.gov NCT04723160

npj Digital Medicine (2024)7:8; <https://doi.org/10.1038/s41746-023-00991-9>

INTRODUCTION

Fundus diseases have become the most common irreversible leading causes of blindness worldwide¹. With the increase in the

image recognition, the specialty of ophthalmology has become particularly well suited to DL techniques and their real-world application². Over the last decades, advances in DL technology

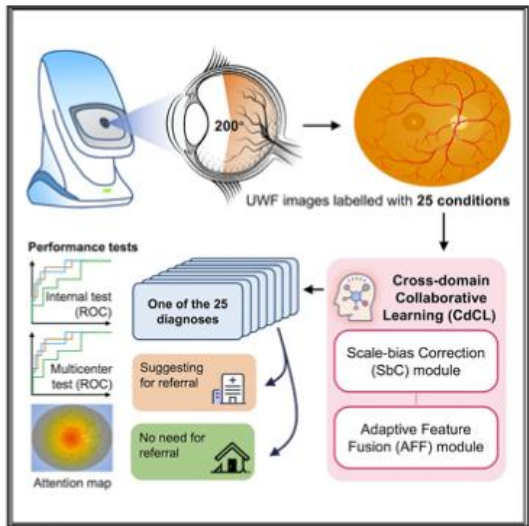
An example of a paper published in Cell Reports- Capable of diagnosing 25 diseases by analyzing the ultra-wide-field fundus image

Cell Reports
Medicine

Article

Ultra-wide-field fundus photography and AI-based screening and referral for multiple ocular fundus diseases

Graphical abstract



Authors

Xinyu Zhao, Xingwang Gu, Da Teng, ..., Qiuming Li, Dawa Xiao, Youxin Chen

Correspondence

chenyx@pumch.cn

In brief

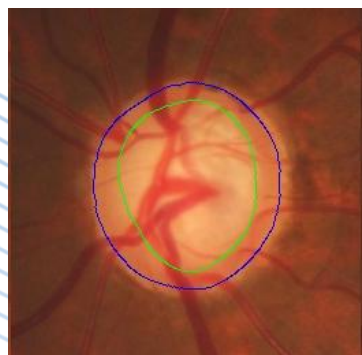
Zhao and Gu et al. develop a deep learning algorithm based on UWF fundus images to screen for up to 25 ocular fundus conditions. Their study demonstrates the importance of the peripheral retinal region in fundus screening and highlights the potential of AI-based methods for large-scale screening applications.

Highlights

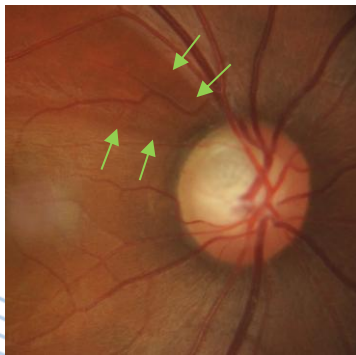
- The WARM model can accurately screen for up to 25 fundus conditions
- The model is trained and tested using multi-center data collected across China
- UWF fundus imaging enhances the model's ability to detect peripheral lesions

» EyeWisdom Glaucoma

The value of early screening and early diagnosis of glaucoma: the world 's first irreversible blinding eye disease, early symptoms hidden. Product innovation: 1) Auxiliary diagnosis suggestions combined with multiple optic nerve characteristics; 2) Constructing an automatic analysis model of retinal nerve fiber layer defects (RNFLD); 3) Constructing an automatic detection model of "optic disc hemorrhage"; 4) Describe the characteristics of the cup and disc based on the contour of the optic disc; The product **have passed NMPA' s Special Examination and Approval Procedures for Innovative Medical Products**,



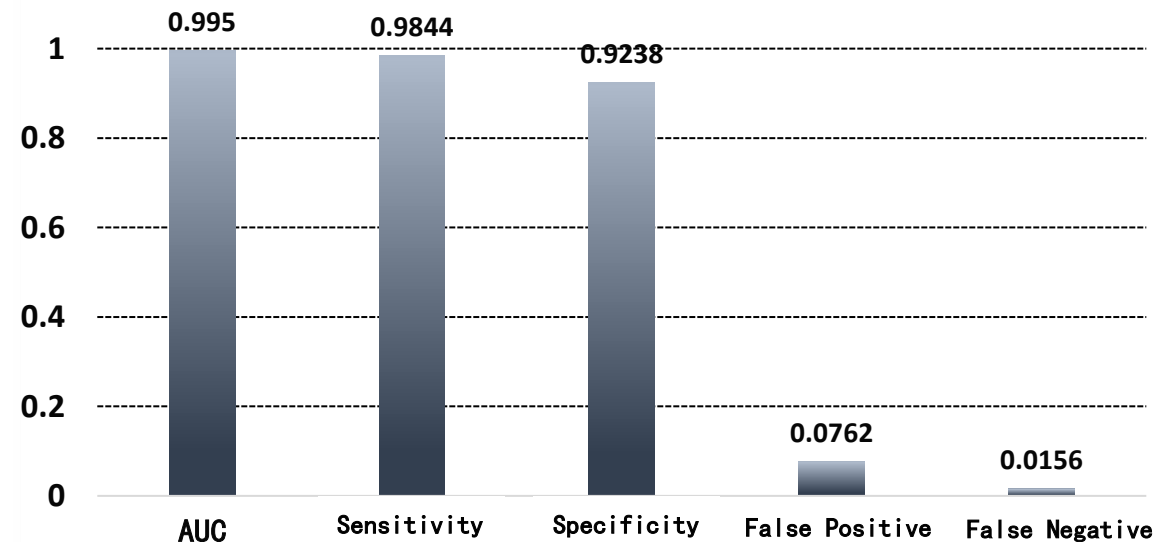
Cup/Disc Segmentation



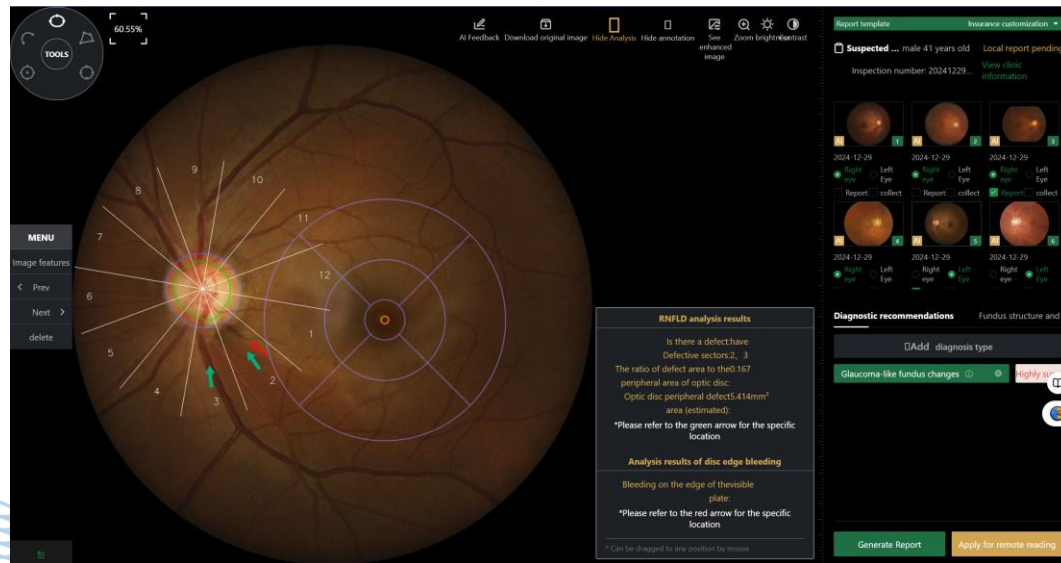
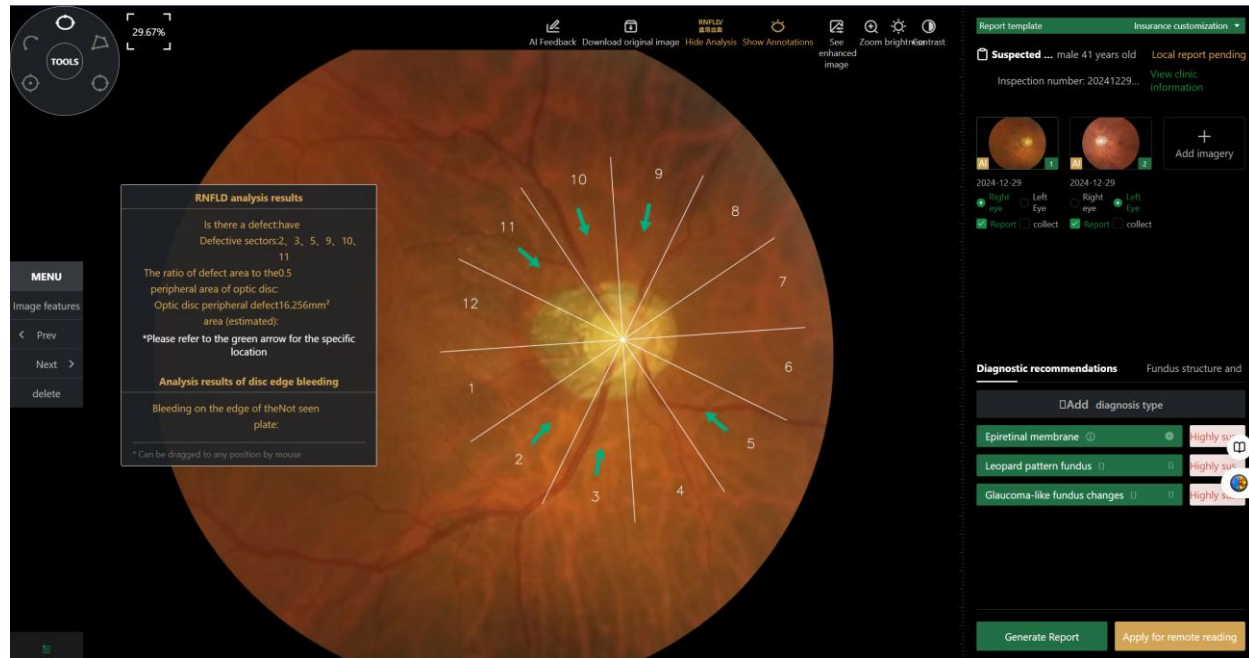
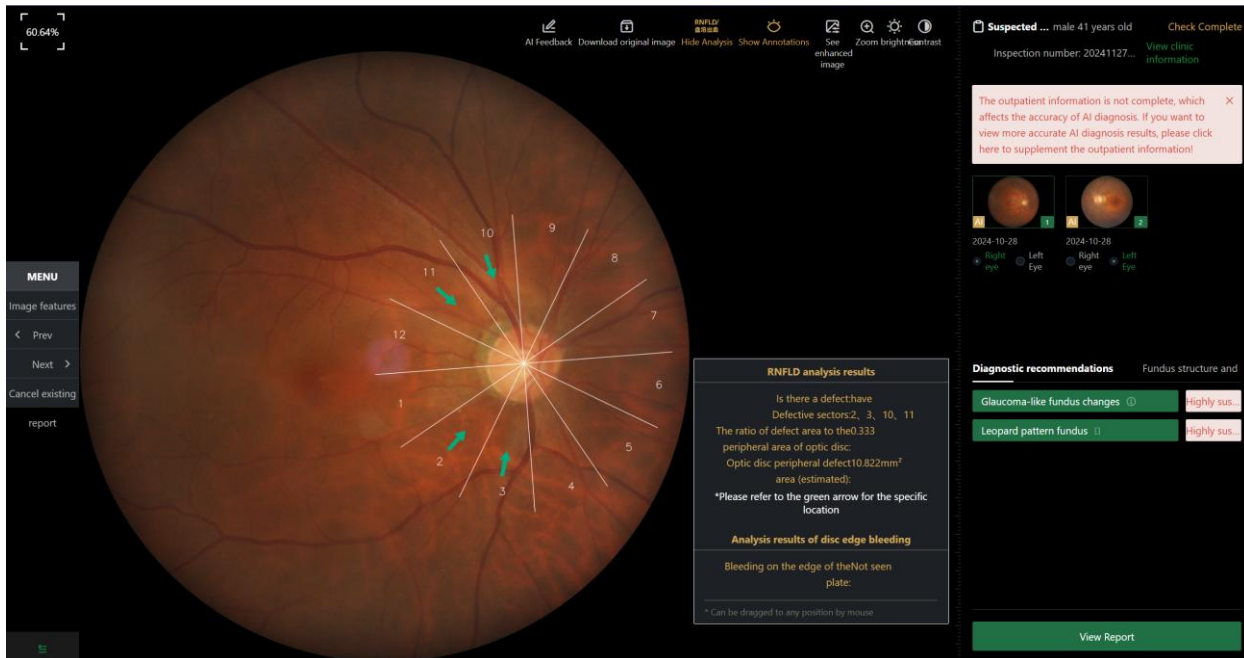
RNFLD Detection



Optic Disc hemorrhage Detection



EyeWisdom Glaucoma Demo



Automatically Generated Report which can be customized

Demo Hospital

Patient Information

Name

AMD

Patient ID

20241205002

Date of Birth

1968-12-05

Gender

Male

Retinal Image Assessment and Management Plan

Fundus Photograph

Right eye (OD)

Left eye (OS)

Screening Summary

Right eye (OD)

C/D Ratio ≈ 0.6

no DR

glucocorticoid induced cataract

highly suspected

tigroid

highly suspected

nonexudative age-related macular degeneration

highly suspected

Left eye (OS)

C/D Ratio ≈ 0.4

no DR

nonexudative age-related macular degeneration

highly suspected

Recommendation and Management Plan

In the event of diminished visual acuity or ocular discomfort, prompt referral to an ophthalmologist for further evaluation is recommended to ascertain and mitigate any potential risks.

Recommended for re-examination every six months.

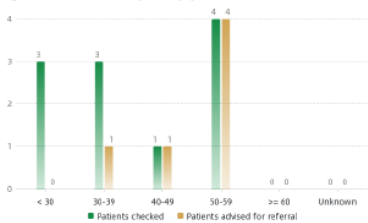
It is advisable to refer to an ophthalmologist, The result of the AI software shows suspect/risk of anomalies in the portion of the posterior pole analyzed. Periodic appointments with an ophthalmologist helps to prevent retinal diseases that might cause vision damages. The indications given are partial and provided by an artificial intelligence software. Artificial intelligence software analysis is not a replacement of a diagnosis made by an ophthalmologist but it is intended only to retrieve particular features in the portion of the fundus examined. The software processes pictures taken on limited portion of retina. Even if anomalous features are not retrieved by the software, this does not guarantee the absence of the disease.

Rich Statistics Data

AI Screening

EyeWisdom AI	Checked Patients 11	Images 31	Reports 7	Remote Reports 0
Demo Hospital	No obvious abnormalities(?) 2	Abnormal Fundus(?) 9	Recommend for referral 6	Poor quality images 0
Ophthalmology	Statistical Range 2024-11-24 to 2024-12-24	18.2% 81.8%	54.5%	0% Export Time: 2024-12-25

Age distribution of the inspected population



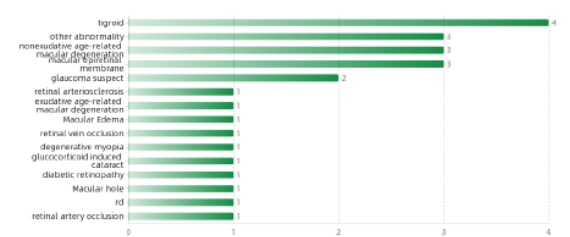
Gender distribution of the population recommended for referral



Gender distribution of all inspected population

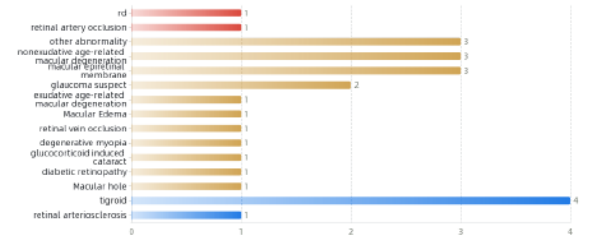


Detection results of fundus diseases



Statistics explanation:
The bar chart here shows the number of people who have been identified as possibly having the above diseases within the selected time range.
Since one person may have multiple diseases, the total number of people with a single disease in the bar chart is inconsistent with the result of "the number of people with clear abnormalities", which is normal.

Severity statistics of fundus diseases



Critical Abnormal
11.1%

It has a great impact on the health of the fundus and develops rapidly. It needs to be referred to a specialist hospital for treatment immediately. If not handled in time, it will cause a serious decline in vision or even blindness.

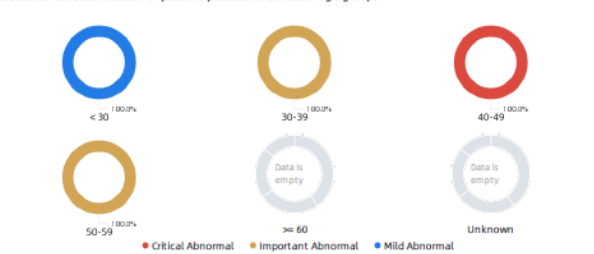
Important Abnormal
55.6%

It has a great impact on the health of the fundus, but the development speed is not as fast as "critical diseases". If it is not controlled and treated, it will cause serious damage to the vision as the disease progresses. You should be reminded to consult a specialist hospital for treatment options and have regular fundus examinations in daily life.

Mild Abnormal
33.3%

The impact on the health of the fundus is not obvious for the time being. Check the fundus regularly and pay attention to changes in vision. If you feel uncomfortable, seek medical attention in time.

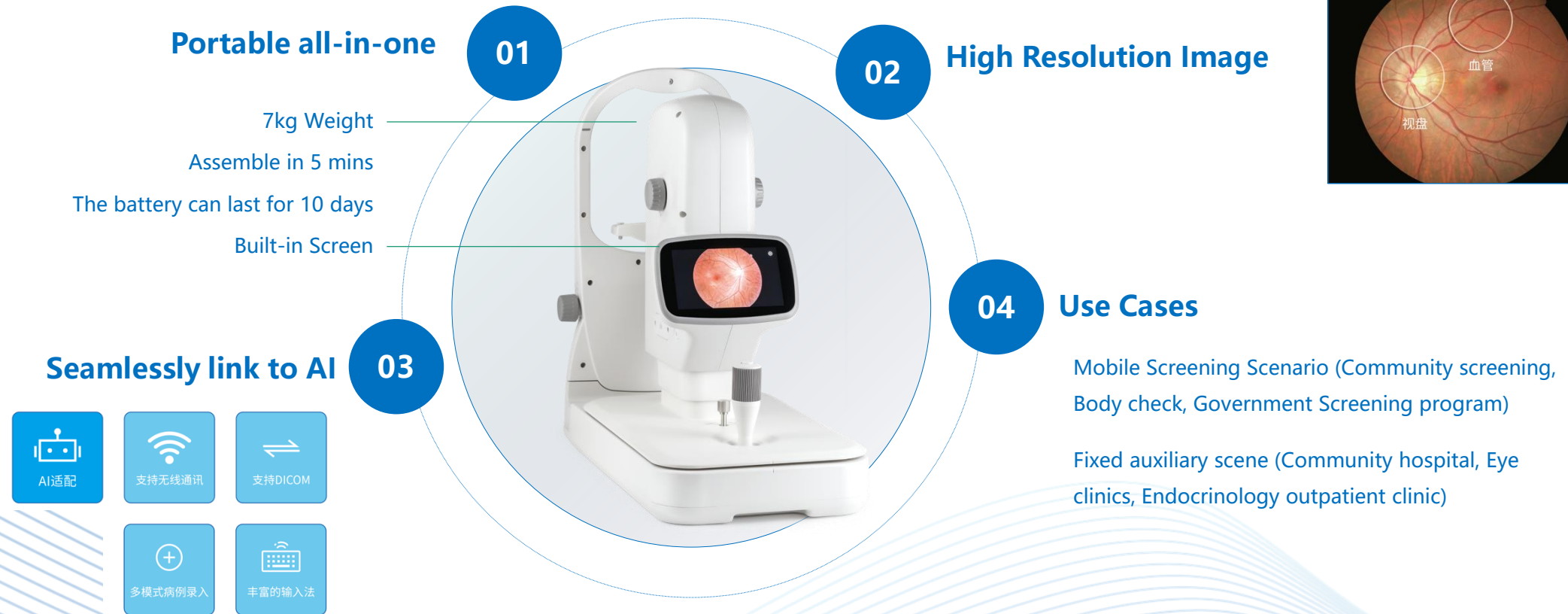
Detection rate distribution of positive patients in different age groups



Explanation and statistics of each disease



➤ RC350 Portable Cost-effective Fundus Camera (no dilation)



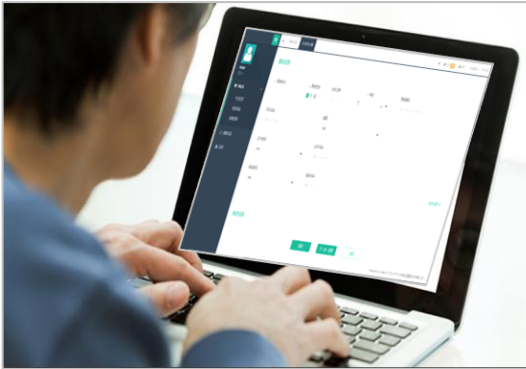
➤ RX350 Fully Automatic Fundus Camera



» The Step-by-Step AI Services

1

Screening preparation: The GP / nurse gather patient information, medical history.



3

GP / nurse send the data to the AI cloud service



If network is NOT an issue, the AI software takes 10 seconds to generate the result.

2

The GP / nurse take fundus photos for the patient.



4

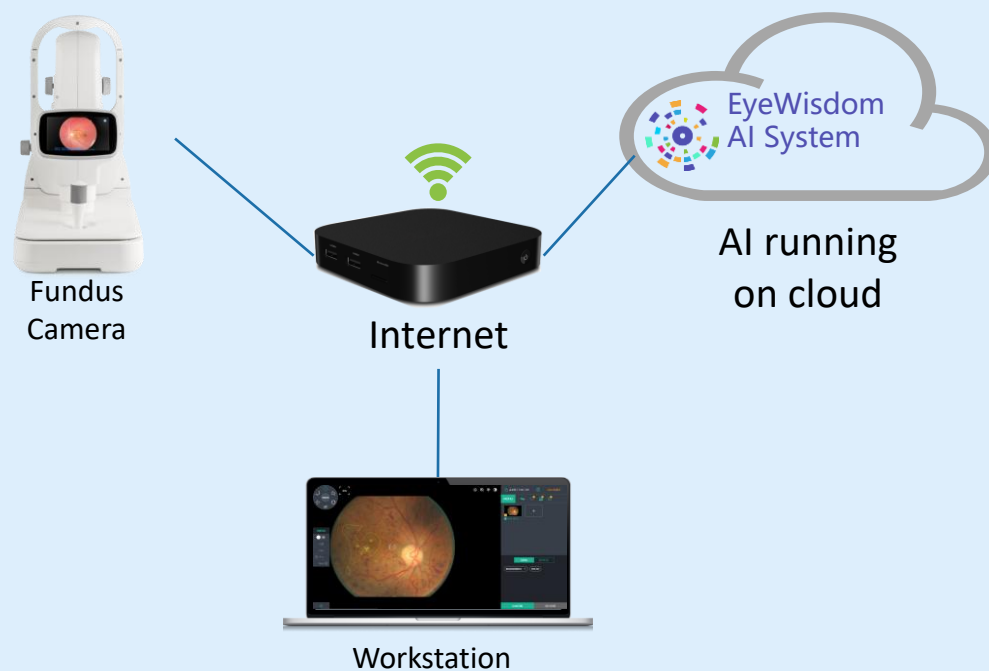
AI result generated, and GP / nurse download the diagnosis and send it back to patient.



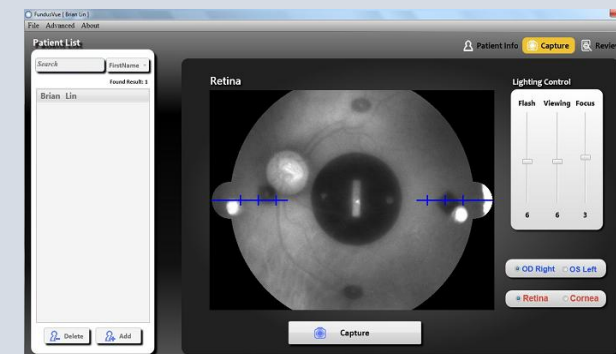
Vistel

» Fundus Camera & System

■ ImageHub: Encryption data transmission



■ Typical specs of Fundus Camera



- 45 degree color / no red light image
- LED light source
- Front LED lighting
- -35D-+30D refractive correction
- Image sharpening algorithm
- 5.2 megapixel image storage

>> Who should take eye exams?



AAO Recommendation:

The recommended frequency for adult comprehensive medical eye examinations for asymptomatic patients, and for patients who do not have risk factors for eye disease, is as follows: under 40 years--every 5 to 10 years; 40 to 54 years--every 2 to 4 years; 55 to 64 years--every 1 to 3 years; and 65 years or older--every 1 to 2 years.

For young individuals at higher risk for certain diseases, such as African-Americans who are at higher risk for glaucoma, comprehensive eye examinations should be considered every 2 to 4 years for those under age 40, every 1 to 3 years for those aged 40 to 54, and every 1 to 2 years for those aged 55 to 64, even in the absence of visual or ocular symptoms.

Individuals who develop diabetes mellitus type 1 should be examined by an ophthalmologist 5 years after disease onset and at least yearly thereafter. Individuals who develop diabetes mellitus type 2 should be examined at the time of diagnosis and at least yearly thereafter. Women with type 1 or type 2 diabetes should receive a comprehensive eye examination before conception and then early in the first trimester of pregnancy. Recommended intervals for subsequent examinations depend upon the level of retinopathy.

<https://www.aao.org/education/clinical-statement/frequency-of-ocular-examination#:~:text=Individuals%20who%20do%20not%20have%20symptoms%20but%20who,of%20eye%20disease%20require%20periodic%20comprehensive%20eye%20examinations.>



AOA Recommendation:

Recommended Eye Examination Frequency for Adult Patients**

Examination Interval

Patient Age (Years)	Asymptomatic/ Low-Risk	At-Risk
18 through 39	At least every two years	At least annually, or as recommended
40 through 64	At least every two years	At least annually, or as recommended
65 and older	Annually	At least annually, or as recommended

Source: Comprehensive Adult Eye and Vision Examination



➤ Where are the patients at in public / private healthcare system?

- Endocrinology where diabetic patients are the main patients.
- GP clinics / entry level hospitals.
- Physical Examination Center / health management center
- Government driven screening programs

» Applications in China

1000+

Medical institutions daily use

3Million+

Patients served

2022CIFTIS



Vision Health Comprehensive
Management Program
Shanghai City



Diabetics Management Program
Meishan City, Sichuan Province



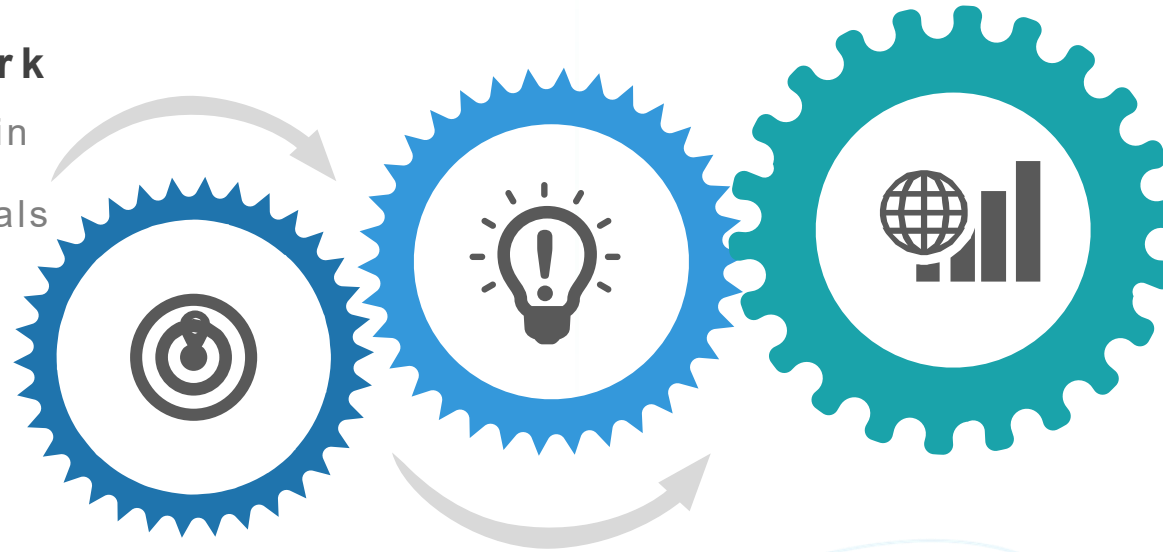
"Eye Health Care Empowered by AI"
Special Campaign
Gulou District, Xuzhou City,
Jiangsu Province

» Shanghai, China

The Shanghai Eye Disease Prevention & Treatment Center applies artificial intelligence to effectively organize and strengthen the tiered diagnosis and treatment system to bring services for eye health and wellbeing closer to large communities.

Three-tiered eye health network

Hardware+AI deployed in
100+ community hospitals



Internet-based remote management

helped
15000+ patients
avoid the risk of blindness

Image collection

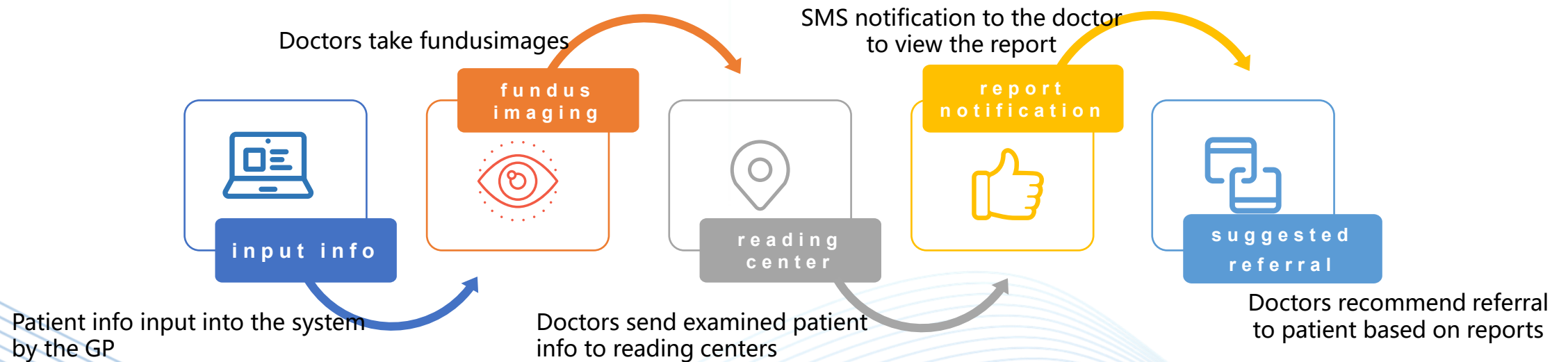
40000+ AI-based auxiliary diagnoses of
multiple fundus diseases for above-60s

➤ Raffles Medical, Beijing Clinic



RafflesMedical
Now Serving You in 13 Cities in Asia

"AI screening+Remote image-reading" Solution



Poland: National DR Screening Program

Vistel Community Hospital

Screening Report

Patient Information

Name

Jane Smith

Patient ID

8009003

Date of Birth

6-16-1963

Gender

Female

Retinal Image Assessment and Management Plan

Right eye (OD)

Left eye (OS)

Fundus Photograph

Screening Summary

Right eye (OD)

Left eye (OS)

C/D Ratio = 0.8 ↑

C/D Ratio = 0.4

Nonexudative age-related macular degeneration

Myelinated retinal nerve fiber

Hypertensive retinopathy

Hypertensive retinopathy

Retinal arteriosclerosis

Recommendation and Management Plan

• If symptoms such as a reduction in visual acuity or other ocular discomfort are experienced, an immediate, comprehensive examination at the ophthalmology department is strongly recommended to identify or exclude potential risks.

• Periodic reviews, ideally every 3 to 6 months are advised.

• Additionally, it's imperative to monitor systemic conditions regularly, including blood pressure, blood sugar, and lipid profiles. Regular follow-up visits should be maintained for optimal health management.

Reviewing Physician: Dr.Lauren

Report date: 2023-03-19

It is advisable to refer to an ophthalmologist. The result of the AI software shows suspicion of anomalies in the portion of the posterior pole analyzed. Periodic appointments with an ophthalmologist helps to prevent retinal diseases that might cause vision damages. The indications given are partial and provided by an artificial intelligence software. Artificial intelligence software analysis is not a replacement of a diagnosis made by an ophthalmologist but it is intended only to relieve particular features in the portion of the fundus examined. The software processes pictures taken limited portion of retina. Even if anomalous features are not retrieved by the software, this does not guarantee the absence of the disease.



Wyniki wykrywania chorób dna oka

16 sites across the country

Diabetic program at GPs in Italy: data and results

Objective: demonstrate that the screening and follow-up model for diabetic retinopathy based on the collaboration between the GPs and the hospital is well accepted and increases the level of healthcare service.

POPULATION

231 diabetic patients had their **retinography** at GP cabinets with Nexy without reaching the public hospital (ULSS 6 – Schiavonia (PD), Italy) for their mandatory DR follow-up.

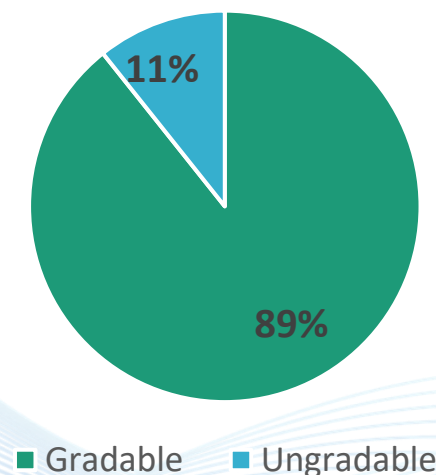
DATA

Retinal images are processed with AI and shared with the remote ophthalmologists thanks to Nexy DICOM connectivity.

- 467 images in the initial dataset
- 417 images are **gradable** for the AI (quality level 1-4) **(89%)**
- 274 images have a DR grading and were used as dataset to test the accuracy of the solution
- **Sensitivity = 100% and specificity=97,7%**

		AI outcome	
		<i>referrable</i>	<i>un-referrable</i>
Ground truth	<i>Referrable</i>	11	0
	<i>un-referrable</i>	6	257

AI: IS THE DATASET GRADABLE?



* «referrable» is mtmDR (more than mild DR)

** «un-referrable» is no DR + mild DR (both conditions require 1 year follow-up in clinical practice)

AI in diabetic clinics in Italy: data and results

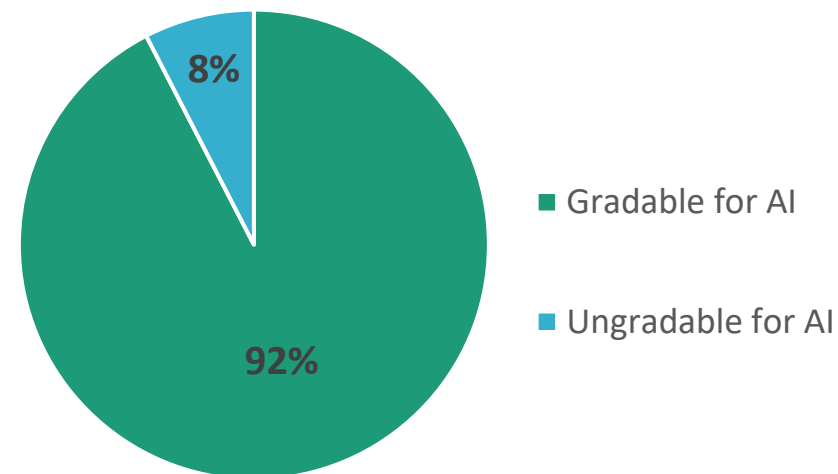
DATA :

- total **198 images**
- remotely reviewed by 2+1 ophthalmologists in masked fashion to each other and to AI outputs
- ground truth is defined as consensus of 2 over 3 doctors among *no referable DR* and *referable DR*
- 183 images are **gradable** for the AI (quality level 1-4) (**92%**)
- 2 over 3 **agreement by doctors** is reached on the **81%** of the dataset
- **Sensitivity = 100%** and **specificity = 88%**

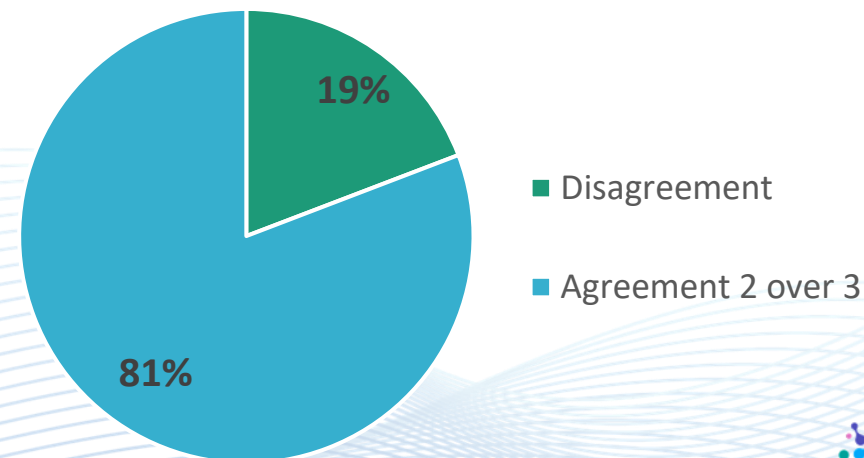
		AI outcome	
		<i>referable</i>	<i>un-referable</i>
Ground truth	<i>referable</i>	5	0
	<i>un-referable</i>	17	121

Note: pictures labeled by the doctor as “Other abnormalities” were considered as “no referable DR” since they were no DR-related.

AI: IS THE DATASET GRADABLE?



Ground truth dataset



Pilot Launch of Artificial Intelligence Ophthalmology Diagnosis Solution



Objectives

- Accuracy Validation: Validate diagnostic accuracy of Vistel AI solution vs. usual methods
 - User Feedback: Collect qualitative feedback on usability and effectiveness.
 - Integration Assessment: seamless integration into existing clinical workflows, reports, and statistical analyses.
-
- Identify any challenges and barriers to implementation.
 - Measure the impact of AI diagnosis on patient's health plans, and satisfaction.

Methodology

Training

1. Conduct comprehensive training sessions for healthcare professionals at each launch location.
2. Provide hands-on experience with the AI system, including troubleshooting and user interface navigation.

Data Collection

1. **Quantitative Data:** Diagnostic accuracy rates, time taken for diagnosis, and number of patients evaluated.
2. **Qualitative Data:** Surveys and interviews with healthcare professionals regarding their experience and observations.

Expected Outcomes

- Achieve a diagnostic accuracy rate of at least 90% compared to traditional methods.
- Reduce the average time for diagnosis and improve reporting and workflow efficiency.
- Increase healthcare professionals' confidence in their diagnosis and speed with an AI diagnostics assistance.
- Positive patient feedback regarding the speed and accuracy of diagnosis.